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ECG Anomaly Detection using Variational Autoencoders

1. Introduction

Electrocardiogram (ECG) signals play a critical role in monitoring heart activity and identifying abnormalities such as arrhythmias. However, manual analysis of ECG signals is time-consuming and requires domain expertise, making automated solutions essential in modern healthcare systems.

This project focuses on building an unsupervised anomaly detection system using a Variational Autoencoder (VAE). Unlike traditional supervised models, the VAE learns patterns from normal ECG signals and detects anomalies based on reconstruction error. This approach is especially useful in real-world scenarios where labeled abnormal data is limited or unavailable.

2. Problem Statement

Healthcare systems require reliable and scalable solutions to detect abnormal heart patterns in real time. Traditional rule-based or supervised approaches often struggle due to:

- Limited labeled anomaly data
- High variability in ECG patterns
- Complexity of time-series signals

This project aims to answer:

- How can deep learning models learn normal ECG behavior effectively?
- Can reconstruction-based approaches identify anomalies without explicit labels?
- How well does a VAE perform in distinguishing normal vs abnormal ECG signals?

3. Dataset and Preprocessing

Dataset Overview:

The ECG5000 dataset consists of time-series ECG signals categorized into multiple heartbeat classes. For this project:

The dataset is converted into a binary classification problem:

- Normal signals
- Anomalous signals

Data Preprocessing Steps:

- **Data Merging**: Train and test datasets were combined for consistency
- **Label Transformation**: Multi-class labels converted into binary labels
- **Normalization**: Input signals scaled for stable model training
- **Training Strategy**:
  - Model trained only on normal ECG signals
  - Both normal and anomaly data used during evaluation

This setup ensures that the model learns only healthy patterns, making anomalies easier to detect.

4. Model and Approach

Why Variational Autoencoder (VAE)?

The VAE is a deep learning model designed to learn compressed representations of input data while preserving important patterns.

It consists of:

- **Encoder** → Compresses ECG signals into latent space
- **Latent Space** → Represents learned distribution of normal signals
- **Decoder** → Reconstructs the original signal

Core Idea: The model learns to reconstruct normal ECG signals accurately.

When an anomalous signal is passed:

- Reconstruction error increases
- Model fails to reproduce the pattern correctly

This difference is used for anomaly detection.

5. Model Training Strategy

- The model is trained only on normal ECG signals
- Loss function includes:
    - **Reconstruction Loss (MSE)** → Measures reconstruction accuracy
    - **KL Divergence** → Regularizes latent space
- Training is performed over multiple epochs to:
    - Minimize reconstruction loss
    - Learn stable latent representations

This ensures that the model captures the true distribution of normal heart signals.

6. Results and Evaluation

Anomaly Detection Mechanism:

- Each test sample is passed through the VAE
- Reconstruction error is calculated
- A threshold is defined to classify anomalies

Threshold Selection:

- A threshold based on the 90th percentile of reconstruction error from normal training samples was used to classify anomalies.

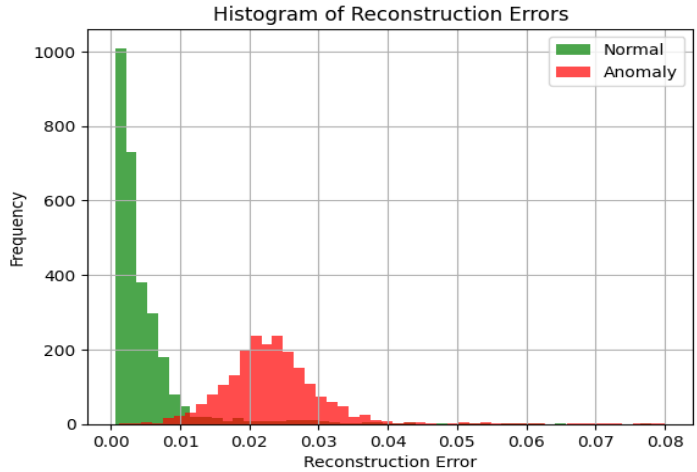
- This approach ensures that the majority of normal signals are correctly reconstructed while signals with significantly higher reconstruction error are identified as anomalies.

Key Evaluation Components:

<u>1. Reconstruction Error Distribution:</u>

- Normal signals → Low error
- Anomalies → High error

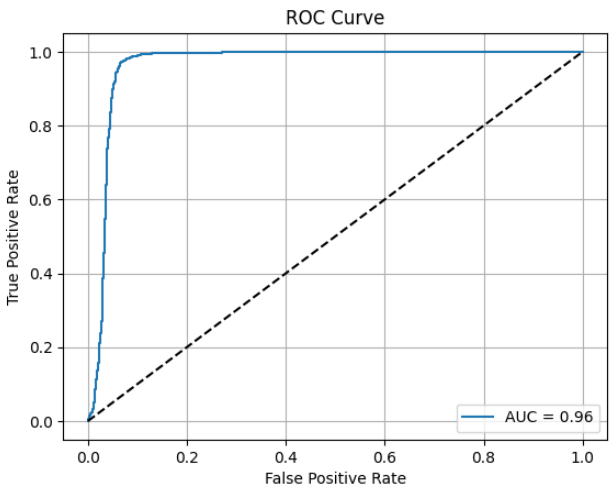
Clear separation indicates strong model performance



Interpretation: The reconstruction error distribution shows a clear separation between normal and anomalous signals. Normal ECG signals have consistently low reconstruction error, while anomalous signals exhibit significantly higher error values. This confirms that the model has effectively learned the pattern of normal signals and can detect deviations reliably.

<u>2. ROC Curve:</u>

- Evaluates classification performance across thresholds
- Higher AUC indicates better separation



Interpretation: The ROC curve demonstrates strong classification performance, with an AUC score close to 1. This indicates that the model can effectively distinguish between normal and anomalous ECG signals across different threshold values, making it a reliable anomaly detection approach.

Model Performance Metrics:

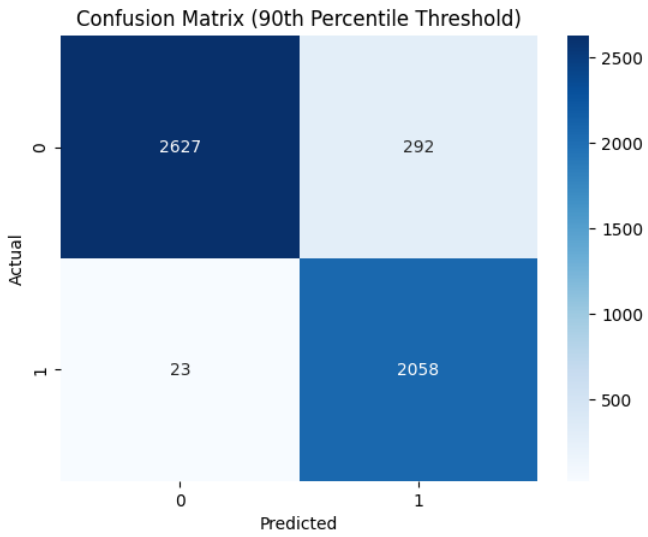
To quantitatively evaluate the model performance, the following metrics were observed:

- **AUC Score:** 0.96
- **Precision:** High precision indicating most predicted anomalies are correct
- **Recall:** High recall indicating most actual anomalies are detected
- **F1-Score:** Balanced performance between precision and recall

These metrics confirm that the model performs effectively in distinguishing between normal and anomalous ECG signals.

3. Confusion Matrix:

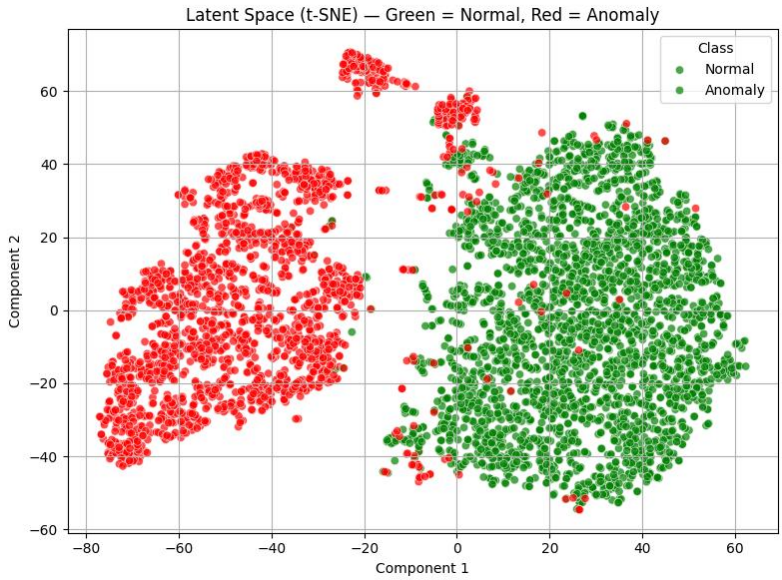
- Shows classification accuracy
- Helps analyze:
    - False positives
    - False negatives



Interpretation: The confusion matrix indicates strong classification performance, with a high number of correctly classified normal and anomalous signals. The relatively low number of false negatives suggests that the model is effective in detecting anomalies, which is particularly important in healthcare applications where missed anomalies can have serious consequences.

4. Latent Space Visualization (t-SNE):

- Visualizes clustering behavior
- Normal and anomaly points form distinct groups



Interpretation: The t-SNE visualization shows clear clustering between normal and anomalous signals in the latent space. This indicates that the VAE has learned meaningful representations, allowing it to separate different patterns effectively.

Overall Result Summary:

The model demonstrates strong anomaly detection capability, with clear separation between normal and anomalous signals, high AUC score, and low false negatives, making it reliable for real-world healthcare applications.

7. Key Insights

- The VAE successfully learns patterns of normal ECG signals
- Reconstruction error provides a **strong signal for anomaly detection**
- Clear separation between normal and anomalous signals is observed
- The model performs well even **without labeled anomaly training data**

8. Business Impact

This project demonstrates strong real-world applications in healthcare:

- **Early Detection of Heart Conditions**
  Enables identification of abnormal ECG patterns before critical events
- **Automated Monitoring Systems**
  Can be integrated into wearable devices and hospital systems
- **Reduced Dependency on Manual Analysis**
  Helps doctors focus on critical cases
- **Scalable Healthcare AI Solutions**
  Suitable for large-scale patient monitoring systems

9. Conclusion

This project successfully demonstrates how deep learning can be used for unsupervised anomaly detection in ECG signals. The Variational Autoencoder effectively learns normal patterns and identifies deviations using reconstruction error.

Despite not using labeled anomaly data during training, the model shows strong performance in distinguishing abnormal signals, making it a practical solution for real-world healthcare applications.

10. Future Work

- Improve model performance with:
  - Hyperparameter tuning
  - Deeper architectures
- Explore advanced models:
  - LSTM Autoencoders
  - Transformer-based models
- Extend to:
  - Real-time ECG monitoring systems
  - Multimodal healthcare data
- Deploy as:
  - Web application or API
  - Integration with wearable devices